

KHAYYUM KHAN PATHAN

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PROFESSIONAL SUMMARY

- Software Engineer with **5+ years** of experience in architecting, designing, developing, and delivering scalable enterprise applications using **Java, Spring Boot, Microservices, and Distributed Systems**.
- Expertise in **solution architecture, high-performance API design, event-driven systems (SOA/EDA), platform modernization, and cloud-native application development**, with a strong focus on scalability, performance, and reliability.
- Proven experience in **technical leadership**, translating business requirements into scalable technical solutions, driving architecture decisions, mentoring engineers, and collaborating with cross-functional teams and business stakeholders.
- Strong background in **performance engineering**, optimizing distributed applications through asynchronous processing, multithreading, Redis caching, and Kubernetes (Amazon EKS) deployments supporting high-volume production workloads.
- Hands-on experience across the complete **Software Development Life Cycle (SDLC)**, including requirements analysis, architecture, development, unit testing, integration testing, debugging, production support, CI/CD, and Agile/Scrum methodologies.
- **AWS Certified Cloud Practitioner** with hands-on experience in AWS, Oracle, Cassandra, Redis, Kafka, GraphQL, and modern software engineering practices.

EDUCATION

Master of Science, Information Technology

University of the Cumberlands, Kentucky

AUG 2022-MAY 2024

GPA: 3.9/4

Bachelor of Technology in Electronics and Communication Engineering

Narasaraopeta Engineering College, Narasaraopeta

JUL 2015-MAY 2019

GPA: 8.07/10

SKILLS

Programming Languages: Java, C, Python, Java 8, Java 11

Web Technologies: Angular, React, JavaScript, Bootstrap, HTML, CSS

Cloud Stack: AWS, Azure

Research Area: Gen AI, C#

Databases: Cassandra, MySQL, Oracle, MongoDB

Operating Systems: Windows, Linux

Tools and Frameworks: Excel, PowerPoint, Word, Drools, SpringBoot, Spring Batch, Git, GraphQL, Apache Kafka, Kubernetes, hibernate, Jenkins, Junit, Mockito, SonarQube, Postman, Jira, Confluence, Kibana, Grafana

ACHIEVEMENTS

Certifications: AWS CLF-C02 (Certified Cloud Practitioner),

AZ-900 (Microsoft Azure Fundamentals)

WORK EXPERIENCE

Java Developer | Infosys (Apple) | Texas, United States

NOV 2024-Present

- Architected and led the development of the Mac Buy Flow Evolution (MacBFE) platform powering the Apple Online Store (AOS) and Mobile Apps, enabling a scalable and extensible purchase experience for the Mac product line.
- Designed the end-to-end pricing aggregation service for configurable Mac products, defining service contracts, data flow, and APIs supporting pricing computation, delta pricing, and dynamic product selection.
- Engineered a platform-level product selection algorithm that determines the optimal buyable configuration based on user-selected options, improving pricing accuracy and purchase eligibility.
- Defined the overall Service-Oriented and Event-Driven Architecture (SOA/EDA), enabling loosely coupled services with improved scalability, resiliency, and maintainability.
- Led the technical design lifecycle by translating product requirements into architecture, high-level design, implementation plans, and engineering deliverables while collaborating closely with product stakeholders.
- Owned architectural decisions for new platform capabilities, conducting design reviews and providing technical direction across multiple engineering teams.
- Spearheaded performance engineering initiatives for Apple's Spring and Fall NPI launches, ensuring runtime services remained resilient under peak production traffic.
- Identified system bottlenecks and redesigned critical execution paths using asynchronous processing, multithreading, batching, and non-blocking I/O, increasing API throughput from approximately 1,000 TPS to over 3,000 TPS.

- Introduced distributed Redis caching strategies that significantly reduced response latency and minimized repeated backend computations.
- Planned production capacity by determining optimal Amazon EKS pod sizing and scaling strategies for production and staging environments.
- Researched Kubernetes leader election mechanisms and designed an event-driven cache coordination strategy for distributed cache synchronization.
- Produced executive-level performance analysis reports and presented architectural recommendations and optimization findings to senior engineering leadership.
- Led the initiative to modernize the Mac product comparison platform by replacing static comparison pages with a dynamic, collection-driven architecture for Mobile Apps.
- Architected the tooling framework and data model that enabled product teams to configure and manage comparison content independently without application code changes.
- Partnered directly with business and product teams during solution design discussions, influencing technical decisions while balancing scalability, maintainability, and business priorities.
- Mentored engineers through design discussions, architecture guidance, and code reviews while delegating work across the team to ensure successful delivery.
- Led Agile ceremonies, coordinated cross-functional execution, tracked project milestones, and communicated delivery status and technical risks to engineering leadership and project management.
- Served as the primary engineering point of contact for cross-platform integrations, collaborating with frontend, infrastructure, and platform teams to onboard new services and resolve complex production issues.
- Designed diagnostic APIs and observability capabilities that accelerated root cause analysis and reduced production incident resolution time while meeting stringent SLA commitments.
- Participated in major Apple product launch events by monitoring live production traffic, performing real-time debugging, and driving rapid incident resolution during critical business windows.
- Applied object-oriented design principles and design patterns, including the Builder Pattern, to create extensible, maintainable, and reusable software components.
- Built, tested, integrated, and maintained high-performance Java/Spring Boot microservices while driving code quality through unit testing, integration testing, debugging, and continuous improvement practices.
- Progressed from an individual contributor to a Lead Engineer by owning architecture, driving engineering strategy, mentoring teams, influencing technical decisions, and delivering highly scalable distributed systems that support mission-critical business operations.

Software Engineer | REIT Services Inc | Texas, United States

JUN 2024-NOV 2024

- Architected and developed cloud-native Java/Spring Boot microservices for an intelligent order fulfillment platform, improving order routing efficiency by **45%** through rule-based decisioning and distributed processing.
- Designed event-driven workflows using Apache Kafka and Redis caching, reducing order processing latency by **35%** while ensuring high availability and fault tolerance.
- Implemented RESTful APIs and optimized Oracle database queries, improving transaction throughput by **40%** and supporting high-volume concurrent requests.
- Leveraged Amazon EKS, Docker, Jenkins, and CI/CD pipelines to automate deployments, achieving **99.9%** service availability and reducing deployment time by **60%**.
- Built comprehensive unit and integration test suites using JUnit, Mockito, and Postman, increasing code coverage to **90%** while improving application reliability and production stability.

Associate Technology L1 | Publicis Sapient (U.S. BANK) | Bangalore, India

AUG 2021-AUG 2022

- Architected GraphQL APIs for internal operations, enhancing navigation and troubleshooting, leading to a 40% efficiency boost.
- Engineered an advanced access framework for payment authorizations with the Drools engine, cutting transaction processing time by 40% and increasing authorization accuracy by 25%.
Conducted rigorous API / Microservices functionality testing using Postman, achieving 95% integration and reliability, while maintaining code integrity and high-test coverage with SonarQube, upholding quality standards at 90%.
- Leveraged Apache Cassandra for 99.9% data availability and high performance. Enhanced system reliability and cut latency by 30% with JMS-based Apache Kafka. Improved functionality and user satisfaction by 25% through seamless integration of Java back-end with Angular front-end.
- Orchestrated AWS managed services integration for 50% performance boost.
- Played a pivotal role in swiftly diagnosing and resolving production issues, fostering team collaboration and driving 30% faster issue resolution.

Project Engineer | WIPRO (Roche Diagnostics) | Bangalore, India

JAN 2021-JUL 2021

- Spearheaded critical microservices development for the RSP platform, enabling seamless device registration and configuration, resulting in a 75% increase in productivity.
- Developed a resilient Java and SpringBoot service for fetching URLs, resulting in a 60% reduction in connection setup time.

- Integrated Apache Kafka for asynchronous messaging, reducing data processing time by 25%, and optimized MongoDB storage, boosting data retrieval speeds by 35%.
- Leveraged Azure for comprehensive log monitoring, reducing issue resolution time for device registration and configuration by 40%.
- Orchestrated comprehensive test suites with JUnit, driving a 50% improvement in RSP platform test coverage.

Project Engineer | WIPRO (ABB) | Bangalore, India

AUG 2019-DEC 2020

- Led initiatives to automate access management, yielding a 40% increase in operational efficiency.
- Crafted an advanced automation solution for real-time server health evaluations, achieving a 95% success rate.
- Implemented automated alerts for critical server conditions, reducing downtime by 50%.
- Implemented data persistence using Oracle SQL Server, optimizing queries and improving database performance by 30%.
- Utilized Jenkins to orchestrate Java upgrades and security patching with a criterion of 90% success threshold.

PROJECTS

Food Delivery Application

JAN 2024-APR 2024

- Developed backend Microservices using Java, Java8, Spring Boot for an end-to-end food delivery enterprise app.
- Engineered microservices architecture with DTO design pattern and Map Strcuts, boosting scalability by 40%.
- Integrated Eureka for service discovery and registration, enhancing ecosystem agility and interoperability, contributing to a 30% reduction in service discovery time.
- Utilized AngularJS for micro frontend development, improving UI data consumption by 25%.
- Performed comprehensive unit testing with JUnit and Mockito, ensuring 95% API reliability, and functional testing via Postman.
- Implemented Docker for 90% consistent, portable containerization.
- Orchestrated AWS managed services integration for 50% performance boost.
- Leveraged MySQL on AWS RDS and MongoDB for 60% better data management across microservices.
- Employed AWS EKS for 40% faster containerized service deployment through manifest files, integrated AWS LoadBalancer for 35% enhanced scalability and performance.
- Implemented Jenkins for CI and SonarQube for 90% code quality adherence, and automated Continuous Deployment on Kubernetes with AWS ArgoCD for 80% efficiency improvement.

Microservices Solutioning Program

AUG 2019-SEP 2019

- Pioneered a Microservices Architecture to implement Ecommerce platform functionality.
- Identified subdomains and lead the team to develop required services using Java, Spingboot.
- Integrated all the microservices to an external server for a centralized configuration which helped the team to demonstrate the real-time updating of the properties of services.

Fingerprint authentication for smart vehicle security using IOT

JAN 2019-APR 2019

- Spearheaded an embedded C program using an ATMEGA16 microcontroller to control the process of vehicle security.
- This application helps owners to register their fingerprints to authenticate their vehicles and prevent them from theft.
- An SMS alert will be sent to the respective owner whenever any unauthorized attempts happen to access the vehicle.

LEADERSHIP AND INVOLVEMENT

- Head of the Hobby Club and Organized IETE workshop at college.
- Volunteer, Registration Committee for Cultural Fest in College.